KVISIMINE APPLIED TO PROBLEMS IN GEOGRAPHICAL INFORMATION SYSTEM

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***Abstract*—Images are highly complex multidimensional signals, with rich and complicated information content in geographical information system. For this reason they are difficult to analyze through a unique automated approach. However a KVISIMINE scheme is helpful for the understanding of image content and data content. In this paper, describes an application K-MEAN clustering algorithm and image information mining for exploration of image information and large volumes data. Geographical Information System, is any system that captures, stores, analyzes, manages, and presents data that are linked to location. Technically, a GIS is a system that includes mapping software and its application to remote sensing, land surveying, aerial photography, mathematics, geography, and tools that can be implemented with GIS software Building a GIS is a fruitful area if one likes the challenge of having difficult technical problems to solve. Some problems have been solved in other technologies such as CAD or database management. However, GIS throws up new demands, therefore requiring new solutions. In this paper we have examine difficult problems, and to be solved and gives some indication of the state of the art of current solutions.**

Keywords- K-MEAN Clustering, Visimine, GIS, S-PLUS

I. INTRODUCTION

The purpose of this paper is to present K-MEAN clustering algorithm is to discover internal structure in some set of data points and Color-Based Segmentation Using K-Means Clustering. Our goal is to use this algorithm to segment colors in an automated fashion and supply the points and the number of clusters you expect to get, and the algorithm returns the same points, organized into clusters by proximity. And k-mean algorithm is a non-hierarchical approach to forming good clusters is to specify a desired number of clusters, say, k, then assign each case (object) to one of k clusters so as to minimize a measure of dispersion within the clusters. This approach has become very popular among the bioinformatics crowd, and especially among analysts of gene expression data. And the Visimine system provides the infrastructure and methodology required for the analysis of satellite images. In order to facilitate the analysis of large amounts of image data, we extract features of the images. Large images are partitioned into a number of smaller, more manageable image tiles. Visimine uses an SQL-like query language that enables specification of the data mining task, features to be used in the mining process, and any additional constraints. Visimine data can be accessed from within S-PLUS by using Java connectivity for images and ODBC connectivity for image and region data. In addition, Visimine has the S-PLUS command tool, which provides for easy transfer of images to S-PLUS, and for data processing using the S-PLUS language. The S-PLUS images can be returned to Visimine and displayed there. The rich statistical functionality of S-PLUS, together with the Visimine user interface and the scalability of its data mining engine, allows for easy and powerful customization of the data analysis process. The subject of Geographical Information Systems has moved a long way from the time when it was thought to be concerned only with digital mapping. Whereas digital mapping is limited to solving problems in cartography, GIS is much more concerned with the modeling, analysis and management of geographically related resources. However, there is a widespread lack of awareness as to the true potential of GIS systems in the future. When the necessary education has been completed, will the systems be there to handle the challenge? It has to be said that the perfect GIS system has not yet been developed. Today's database technology is barely up to the task of allowing the handling of geographic data by large numbers of users with adequate performance. Serious questions have been raised as to whether the most popular form of database, the relational model, will be able to handle the geometric data with adequate response. Certainly, if this data is accessed via the approved route of SQL calls, the achievable speed is orders of magnitude less than that which can be achieved by a model structure built for the task. It is a common problem with systems that contain parts that are front ended by different languages that it is not possible to integrate them properly. Modern query languages such as SQL are not sufficient in either performance or sophistication for much of the major development required in a GIS system - but then one would argue that they were not intended for this. A problem which has to be addressed is spatial queries within the language, since trying to achieve this with the standard set of predicates provided is extremely difficult and clumsy. An example of a spatial query is to select objects "inside" a given polygon. If the route adopted is to provide two databases in parallel, a commercial one driven by SQL and a geometry database to hold the graphics, and then there is a problem constructing queries that address both databases.

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The rest of the paper is organized as follows: Section 2 gives the details of related work, proposed work introduced in section 3.And Preliminaries in Section 4, and we discuss our experiments and the results in section 5. Conclusions are presented in Section 6.

II. RELATED WORK

A great deal of research has been focused the use *o*f GIS in the spatial analysis of an archaeological cave site[1], according to HOLLEY MOYES archaeologists traditionally have viewed geographic information systems (GIS) as a tool for the investigation of large regions, its flexibility allows it to be used in non-traditional settings such as caves. This study demonstrates the utility of GIS as a tool for data display, visualization, exploration, and generation. Clustering of artifacts was accomplished by combining GIS technology with a K-means clustering analysis, and basic GIS functions were used to evaluate distances of artifact clusters to morphological features of the cave. The use of GIS in Archaeological Settlement Research Facts, Problems and Challenges [2], Frankfurt Germany, September 26th 2008 using Free and Open Source Software (FOSS) licenses generally allow free deployment anywhere and for any purpose. No redundant licensing costs, more flexible investment options, full control over development. Stable and long-lived data formats, free and *open standards* instead of "industry" no pressure to deprecate older software or data formats. The current pool of available FOSS is gigantic and growing rapidly. Open source licenses generally allow free deployment anywhere and for any purpose. Clustering With GIS [3], Ece AKSOY, Turkey, presented there is no universally applicable clustering technique in discovering the variety of structures display in data sets. Also, a single algorithm or approach is not adequate to solve every clustering problem. This study aims comparing different software in non-spatial and spatial clustering techniques, which can be used for different aims such as forming regional politics, constructing statistical integrity or analyzing distribution of funds, in GIS environment and putting forward the facilitative usage of GIS in regional and statistical studies. Self Organizing Maps (SOM) algorithm which is the best and most common spatial clustering algorithm in recent years. Geospatial Information and Geographic Information Systems (GIS): Current Issues and Future Challenges in June 8, 2009[4], according to Peter Folger**,** Geospatial information is data referenced to a place a set of geographic coordinates which can often be gathered, manipulated, and displayed in real time. A Geographic Information System (GIS) is a computer system capable of capturing, storing, analyzing, and displaying geographically referenced information. Global Positioning System (GPS) data and their integration with digital maps have led to the popular hand-held or dashboard navigation devices used daily by millions.Challenges to coordinating how geospatial data are acquired and used collecting duplicative data sets. Implementation of the Extended Fuzzy C-Means Algorithm in Geographic Information Systems [5], Ferdinand Di Martino, Salvatore Sessa, in 2009, focused on density cluster methods have elevated computational complexity and are used in spatial analysis for the determination of impact areas. We propose the extended fuzzy c-means (EFCM) algorithm like alternative method because it has three advantages: robustness to noise and outliers, linear computational complexity and automatic determination of the optimal number of clusters. We can use the EFCM algorithm in spatial analysis for the determination of circular buffer areas. These areas can be considered on the geographic map as a good approximation of classical hotspots. Applications to other frameworks like crime analysis, industrial pollution, etc. shall be tried in future works. Issues of GIS data management [6] 2007, this paper deals with current issues of spatial data modeling and management used by spatial management applications. Paper describes ways of solving this problem. Now we can summarize the problem of the GIS and CAD integration. Because of the different characteristics of the GIS/CAD worlds, firstly there's need to decide for some suitable 3D data model, which could maintain complex and structured data types. This model also must be able to maintain the large-scale 3D models produced by CAD as well as low-scale objects used by GIS.

III. PROPOSED WORK

In this paper we proposed KVISIMINE, is the combination of K-mean clustering and Visimine.

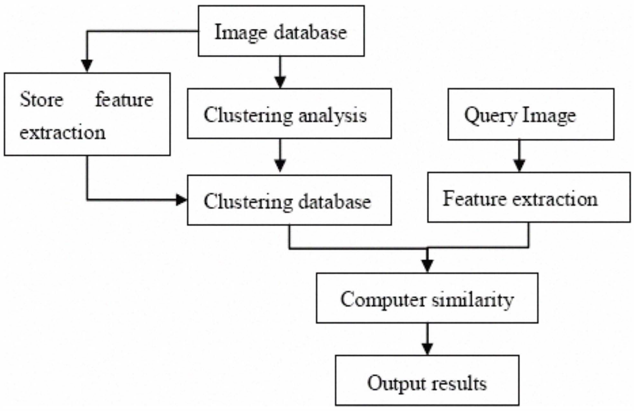

FIG 1 . Combination of K-MEAN and Visimine

A. *K-means clustering*

K-MEANS clustering algorithms, the basic step of k-means clustering is simple. In the beginning we determine number of cluster K and we assume the centroid or center of these clusters. We can take any random objects as the initial centroids or the first K objects in sequence can also serve as the initial centroids. Then the K means algorithm will do the three steps below until convergence Iterate until stable (= no object move group): Determine the centroid coordinate determine the distance of each object to the centroids Group the object based on minimum distance.

The main steps of k-means algorithm are as follows: (1) Select an initial partition with K=k clusters; repeat (2) and (3) until cluster membership stabilizes.

(2) Generate a new partition by assigning each pattern to its

Closest cluster center.(3) Compute new cluster centers.

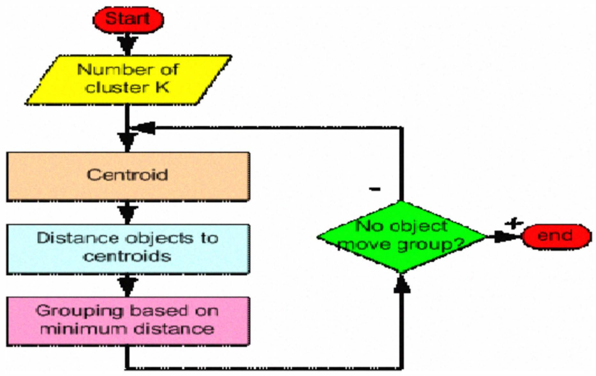

Fig 2 K-MEAN Algorithm

In other word we can say in this algorithm:

- Choose k data points to act as cluster centers
- Until the clustering is satisfactory
  - Assign each data point to the cluster that has the nearest cluster center
  - Ensure each cluster has at least one data point splitting, etc
  - Replace the cluster centers with the means of the elements in the clusters.
  -

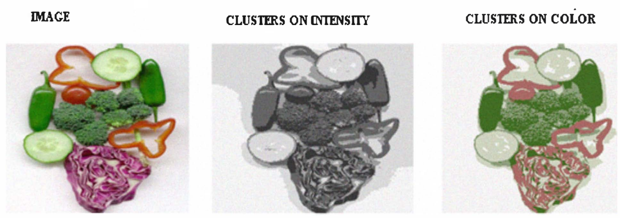

Fig 3 K-MEAN Clustering using Intensity alone and color alone

B. *Database Organization*

Visimine: is a system for data mining and statistical analysis of large collections of remotely sensed images. Visimine system provides the infrastructure and methodology required for the analysis of satellite images. Spatial information about region levels also can be stored in ESRI's Spatial Data Engine (SDE), together with the relevant GIS information. SDE provides open data access across local and wide area networks, and the Internet, using the TCP/IP protocol. This information can be combined with region level features such as texture, spectral properties, or DEM features. The SDE format allows a fusion of GIS, optical, and DEM information for a variety of visualization methods and data analysis functions. The query language allows the user to specify the type of knowledge to be discovered, the set of data relevant to the mining process, and the conditions that have to be satisfied by the data. Based on this query, an SQL statement is constructed to retrieve the relevant data.

The abilities of the data mining engine are enriched through the Java connection to the S-PLUS statistical data analysis engine. The graphical user interface enables browsing and manipulation of the satellite images and associated features, creation of data mining queries, and visualization of the results of the data analysis.

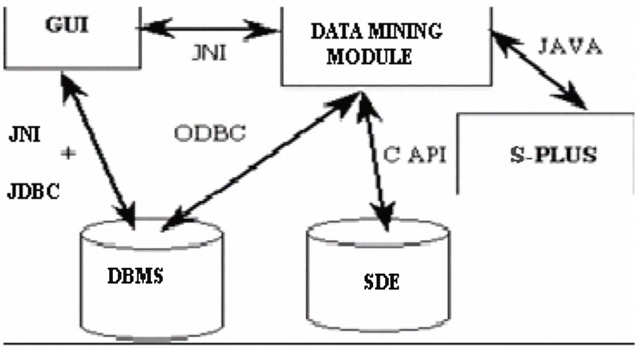

Fig 4 Visimine System

Visimine uses an SQL-like query language that enables specification of the data mining task, features to be used in the mining process, and any additional constraints. The system is capable of performing similarity searches based on any combination of features. In Visimine we use a method for training of land cover labels that employs naïve Bayesian classifiers. Visimine is based on decision tree models. In total S-PLUS contains over 3000 functions for scientific data analysis. Visimine data can be accessed from within S-PLUS by using Java connectivity for images and ODBC connectivity for image and region data. The Visimine can also display S-PLUS graphics, which are created using a command line interface and shown within S-PLUS plot window. The combination of S-PLUS and Visimine features creates a unique environment for interactive exploration and analysis of remotely sensed data.

IV. PRELIMINARIES

Problem definition: Modern query languages such as SQL are not sufficient in either performance or sophistication for much of the major development required in a GIS system - but then one would argue that they were not intended for this. One can see why people like SQL; it can give immense power in return for some fairly simple "select" constructs. A problem which has to be addressed is spatial queries within the language, since trying to achieve this with the standard set of predicates provided is extremely difficult and clumsy. If the route adopted is to provide two databases in parallel, a commercial one driven by SQL and a geometry database to hold the graphics, then there is a problem constructing queries that address both databases. Ideally, the query language should be a natural subset of the front end language allowing access to the same seamless environment that the front end language provides. Much work needs to be done in the area of query languages for GIS.

V. EXPERIMENTAL RESULTS

The KVISIMINE can also display S-PLUS graphics, which are created using a command line interface and shown within S-PLUS plot window. The combination of S-PLUS and Visimine features creates a unique environment for interactive exploration and analysis of remotely sensed data and large images. The rich statistical functionality of S-PLUS, together with the Visimine user interface and the scalability of its data mining engine, allows for easy and powerful customization of the data analysis process. The Visimine provides the infrastructure and methodology required for the analysis of satellite, land, water images. In order to facilitate the analysis of large amounts of image data, we extract features of the images. Large images are partitioned into a number of smaller (segmentation), more manageable image tiles. Partitioning allows fetching of just the relevant tiles when retrieval of only part of the image is requested, and provides faster segmentation of image tiles. Individual image tiles are processed to extract the feature vectors. In fig 5 display partition of image and after partition we perform the cluster of image and store in database and then final display output in S-PLUS in four windows the data as well as graph windows and bar chart this is more helpful in GIS, with the help of S-PLUS we find the area and plot the graph according to requirement, and access images from the database easily

PERFORMANCE EVALUATIONS

S-Plus connectivity: Insightful S-Plus is an interactive computing Environment for graphics, data analysis, statistics, and mathematical computing. It contains a superset of the S object-oriented language and system originally developed at AT&T Bell Laboratories, and it provides an environment for high-interaction graphical analysis of multivariate data, modern statistical methods, data clustering and classification, and mathematical computing. In total, S-Plus contains over 3000 functions for scientific data analysis. Visimine data can be accessed from within S-Plus by using Java connectivity for images and ODBC connectivity for image and region data. In addition, Visimine has the S-Plus command tool, which provides for easy transfer of images to S-Plus, and for data processing using the S-Plus language. The S-Plus images can be returned to KVISIMINE and displayed there. The KVISIMINE display S-Plus graphics, which are created using a command line interface and shown within S-Plus plot window. And output figure shows a graph of the brightness and greenness indices for an agricultural part of British Columbia. The combination of S-Plus and KVISIMINE features creates a unique environment for interactive exploration and analysis of remotely sensed image and data. The rich statistical functionality of S-Plus, together with the approach user interface and the scalability of its data mining engine, allows for easy and powerful customization of the data analysis process.

VI. CONCLUSIONS

In this paper we presented a KVISIMINE S-Plus framework, which explores state-of-the-art data mining and databases technologies to retrieve integrated spectral and spatial information from Geographical information system imagery. A scalable data warehouse containing a huge amount of images may be a better database architecture for fundamentally distributed data management and mining system such as NASA Earth Observing System (EOS). Meanwhile, performance analysis for clustering on and retrieving from large volumes of images is critical for the system to succeed in practical applications. And the results of experiments on the basic of images show that the proposed approach can greatly improve the efficiency and performances of image retrieval, as well as the convergence to user's retrieval concept. Clustering algorithm has been widely used in computer vision such as image segmentation. In addition, current implementation provides data and image based search. A segmentation process can be used to segment an image into non-overlapping regions on which we can further apply the texture feature extraction.

REFERENCES

[1] U. M. Fayyad, G. P. Shapiro, P. Smyth, and R. Uthurusamy (Eds.), Advances in Knowledge Discovery and Data Mining, AAAI/MIT Press, 1996.

[2] M.C. Burl, C. Fowlkes, and J. Roden, "Mining for image content," in Systemic, Cybernetics, and Informatics / Information Systems: Analysis and Synthesis, Orlando, FL, July 1999.

[3] L.-K. Soh and C. Tsatsoulis, "Data mining in remotely sensed images: a general model and an application," in Proceedings of IEEE IGARSS 1998, vol. 2, Seattle, Washington, USA, Jul 1998, pp. 798-800.

[4] J. Zhang, H. Wynne, M. L. Lee, "Image mining: issues, frameworks, and techniques," in Proceedings of 2nd International Workshop on Multimedia Data Mining, San Francisco, USA, Aug 2001, pp. 13 – 20.

[5] G.B.Marchisio andJ.Cornelison,"Content-based search and clustering of remote sensing imagery," in Proceedings of IEEE IGARSS 1999, vol. 1, Hamburg, Germany, Jun 1999, pp. 290 – 292.

[6] A.Vellaikal,C.-C.Kuo, and S. Dao, "Content-based retrieval of remote sensed images uses vector quantization," in Proc. of SPIE Visual Info. Processing IV, vol. 2488, Orlando, USA, Apr 1995, pp. 178 – 189.

[7] R. F. Cromp and W. J. Campbell, "Data mining of multidimensional remotely sensed images," in Proc. 2nd International Conference of Information and Knowledge Management, Arlington, VA, Nov, 1993, pp. 471-480.

[8] Ying Liu, Dengsheng Zhang, Guojun Lu, Wei-Ying Ma.A Survey of content-based image retrieval with high- Level Semantics. Pattern Recognition, Volume 40, Issue 1, January 2007, Pages 262-282.

[9] Muhammad Atif Tahir, Ahmed Bouridane, Fatih Kurugollu. Imultaneous feature selection and feature weighting Using Hybrid Tabu Search/K-nearest neighbor classifier. Pattern Recognition Letters, Volume 28, Issue 4, 1 March 2007.

[10] Sarbast Rasheed, Daniel Stashuk, Mohamed Kamel.Adaptive Fuzzy k-NN classifier for EMG signals Decomposition. Medical Engineering & Physics, Volume 28, Issue 7, September 2006, Pages 694-709.

[11] J. Amores, N. SEbE, P. Radeva.Boosting the distance Estimation: Application to the K-Nearest Neighbor Classifier. Patter

Recognition Letters, Volume 27, Issue 3, February 2006, Pages 201-209.

[12] Man Wang, Zheng-Lin Ye, Yue Wang, Shu-Xun Wang. Dominant sets clustering for image retrieval. M. Wang et al. /Signal Processing 88 (2008) 2843–2849., Venables W. N. and Ripley B. D. (2000), S Programming, Springer, New York..

[13] Craig, N., 2000, Real-time GIS construction and digital data recording of the Jiskairumoko Excavation, Peru: The Society For American Archaeology Bulletin, v. 18, no. 1, p. 24-28.

[14] Edwards, D., 2001, Excavations at Khirbet Cane, Israel, http://anticompetitive/cane.

[15] M.C. Burl, C. Fowlkes, and J. Roden, “Mining for image content,” in Systemic, Cybernetics, and Informatics / Information Systems: Analysis and Synthesis, Orlando, FL, July 1999.

[16] Zlatanova S.: Large-scale data integration An Introduction to the Challenges for CAD and GIS Integration, Directions magazine, July 10, 2004.

[17] Van Ostracism P.: Bridging the Worlds of CAD and GIS, Directions magazine, June 17, 2004.

[18] Zlatanova S., Rah man A. A., Pilouk M.: Trends in 3D GIS Development, Journal of Geospatial Engineering, Vol. 4, No. 2, December 2002.

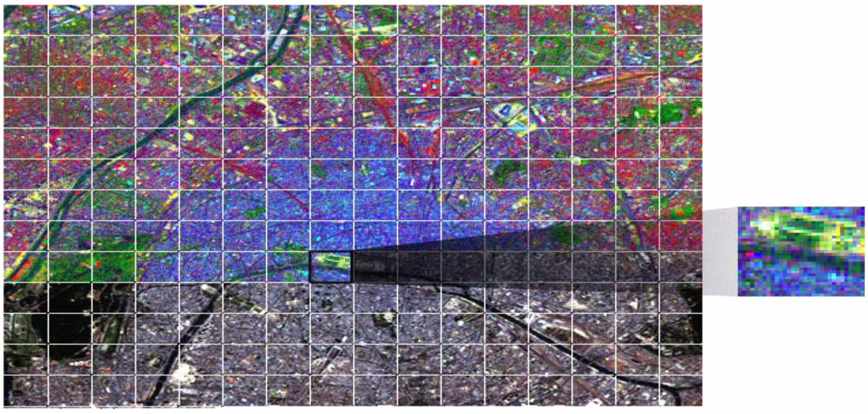

Fig 5 Partition of image

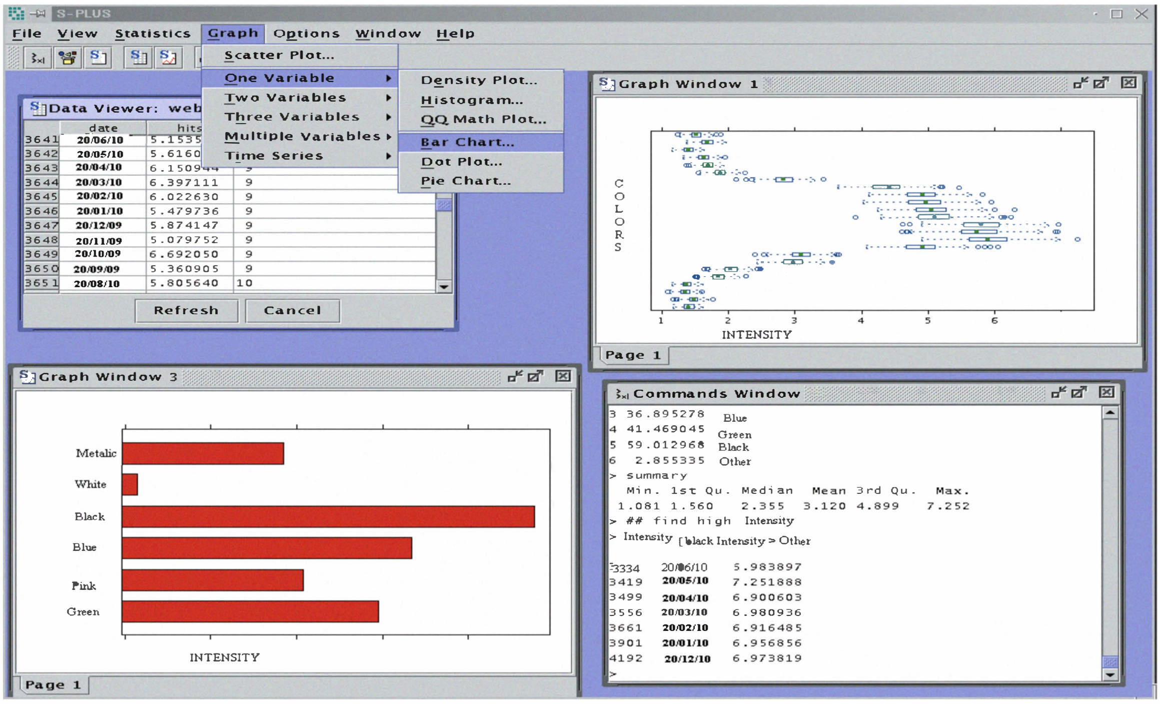

Fig 6 Display Output in S-PLUS.